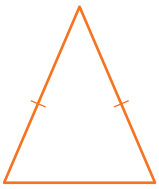
PRIOR LEARNING

Chapter 10: Applications of trigonometry

1 Identify each of the following types of triangles:

a 

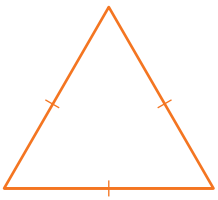
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b 

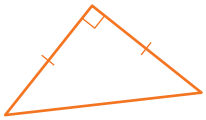
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c 

_____

d 

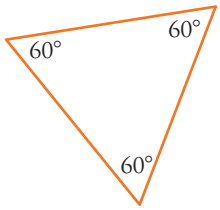
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e 

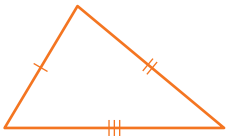
_____

f 

_____

g 

_____

h 

_____

2 Solve:

a $4 = \frac{x}{2}$

b $1 = \frac{x}{5}$

c $7 = \frac{x}{3}$

d $0.6 = \frac{x}{4}$

e $1.8 = \frac{x}{3}$

f $16.2 = \frac{x}{5}$

g $14.7 = \frac{x}{8.2}$

h $108 = \frac{x}{49.2}$

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3 Solve for x, answering to one decimal place where appropriate

a $5 = \frac{10}{x}$

b $3 = \frac{12}{x}$

c $9 = \frac{6}{x}$

d $10 = \frac{7}{x}$

e $4 = \frac{8.3}{x}$

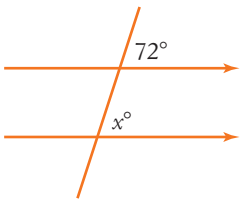
f $2.3 = \frac{7.4}{x}$

g $27.6 = \frac{83.1}{x}$

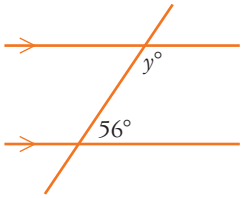
h $96.5 = \frac{52.3}{x}$

4 Find the value of the pronumeral(s) in each diagram, giving reasons for your answers.

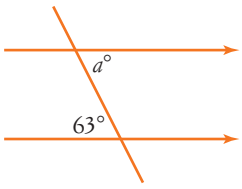
a



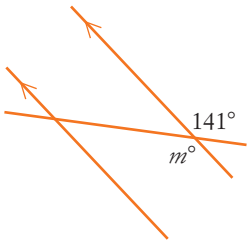
b



c

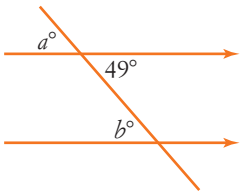


d

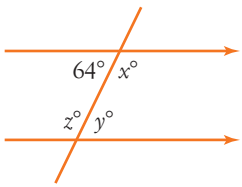


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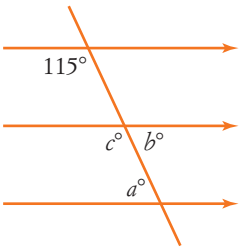
e



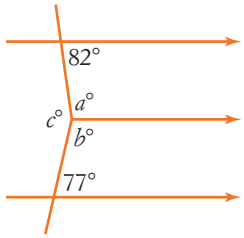
f



g



h



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Answers

1 **a** right-angled
b isosceles
c scalene
d equilateral
e right-angled isosceles
f isosceles
g equilateral
h scalene

2 **a** 8
b 5
c 21
d 2.4
e 5.4
f 81
g 120.54
h 5313.6

3 **a** 2
b 4
c $\frac{2}{3}$
d $\frac{7}{10}$
e 2.1
f 3.2
g 3.0
h 0.5

4 **a** 72° (corresponding angles on parallel lines)
b 124° (co-interior angles on parallel lines)
c 63° (alternate angles on parallel lines)
d 141° (vertically opposite angles)
e $a = 49°$ (vertically opposite angles)
$b = 49°$ (alternate angles on parallel lines)
f $x = 116°$ (supplementary angles)
$y = 64°$ (alternate angles on parallel lines)
$z = 116°$ (alternate angles on parallel lines)
g $a = 65°$ (co-interior angles on parallel lines)
$b = 65°$ (alternate angles on parallel lines)
$c = 115°$ (corresponding angles on parallel lines)
h $a = 98°$(co-interior angles on parallel lines)
$b = 103°$ (co-interior angles on parallel lines)
$c = 159°$ (angles at a point)

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